Electroweak Phase Transition Order from a Microscopic Scalar Rung: A Lean-formalized prediction with EW baryogenesis viability

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The order of the electroweak phase transition (first-order vs. crossover) is a microscopic prediction of the scalar-rung condensate hierarchy on the WetterichNJL substrate. Building on the Wave 1b BHL gauge-embedding formalization (`BHLGaugeEmbedding.lean`, 23 theorems) — itself an extension of the Bardeen-Hill-Lindner / Hill 2025 bilocal mechanism onto the Clifford-16 Fierz basis [1, 2] — we ship `EWPhaseTransition.lean`, 21 theorems formalizing the leading-order high-T effective potential, the first-order / crossover predicate, the latent-heat order parameter, and the EW baryogenesis viability marker. The transition order is determined microscopically by the cubic coefficient E: first-order if $E > 0$ (barrier survives), crossover if $E = 0$.

At the strict-leading-order high-T expansion, the SM benchmark ($m_H = 125.20$ GeV, $\lambda_{\rm SM} \approx 0.13$, $E_{\rm SM} \approx 0.01$) yields $E > 0$ and is therefore predicted first-order; this is the load-bearing Lean theorem `sm_benchmark_is_first_order`. The physical SM, however, is a *crossover*: lattice analyses demonstrate that thermal-loop corrections to E at the physical $m_H = 125.20$ GeV [7] push the effective cubic coefficient below the threshold needed for a first-order transition. The crossover-to-first-order endpoint sits at $m_H \approx 72.4 \pm 1.7$ GeV, refined by the lattice study of Csikor, Fodor, and Heitger [4] (extending the earlier 70–95 GeV range from [3]); the physical m_H is well above this value. The strict-LO Lean prediction (first-order) and the physical lattice verdict (crossover) thus diverge at a known and literature-documented boundary. The Lean theorem `crossover_excludes_baryogenesis` formalizes the consequence: a crossover transition is not viable for sphaleron-decoupling baryogenesis, forcing baryogenesis to leptogenesis or BSM mechanisms — intended to feed the future Phase 6c.2 EW-baryogenesis bridge theorem when that wave activates. Two tracked hypotheses (`H_VacuumStableUnderRG`, `H_HierarchyEWLambdaUV`) encode the SM meta-stability and EW-$_UVhierarchyasfalsifiableparametricconditions; bothhaveexplicitnon-trivialwitnessesandfalsifiers.Thefullformalizationiszero-sorry, introduceszeronewaxioms, andaddszeroheartbeatoverri$

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I. INTRODUCTION

The order of the electroweak phase transition is a quantitative prediction of the SM Higgs sector at finite temperature. The Kajantie-Laine-Rummukainen-Shaposhnikov 1996 lattice analysis [3] established that, *after thermal-loop resummation*, the SM is a crossover for $m_H \gtrsim 70$–95 GeV, with the endpoint refined by Csikor, Fodor, and Heitger [4] to $m_H = 72.4 \pm 1.7$ GeV; with the PDG observed $m_H = 125.20$ GeV [7], the resummed SM transition is a crossover, and EW baryogenesis in the strict-SM (without BSM extensions of the electroweak Higgs sector) is excluded. The strict-LO Lean prediction in this paper instead encodes the un-resummed cubic coefficient and is reframed below as a structural reference point against which the resummed lattice result improves.

The Phase 5z Wave 3 program formalizes this prediction at the level of the WetterichNJL scalar-rung embedding (Wave 1, with BHL gauge extension Wave 1b). The leading-order high-T expansion of the Higgs effective potential takes the form

$$V_T(\Phi, T) = \tfrac{1}{2}(c_T T^2 - \mu^2)\Phi^2 - \tfrac{1}{3} E T \Phi^3 + \tfrac{1}{4}\lambda\Phi^4, \quad (1)$$

where c_T is the thermal-mass coefficient ($\sim (g^2 + g'^2 + 4y_t^2)/16$ in the SM), E is the cubic coefficient from hard-thermal-loop W/Z resummation ($\sim g^3$), and the cubic term is what distinguishes a first-order transition (with a barrier between false and true vacua) from a crossover (smooth interpolation).

This paper formalizes (1) as `EWPhaseTransition.finiteTPotential`, the cubic-coefficient order parameter as `IsFirstOrderEW` / `IsCrossoverEW`, the latent-heat consequence as `latentHeat_nonneg` + `crossover_has_zero_latent_heat` + `first_order_has_positive_latent_heat`, and the baryogenesis viability marker as `IsBaryogenesisViable` + `crossover_excludes_baryogenesis` — in a single Lean module of 26329-substantive-theorem project context.

II. FINITE-T EFFECTIVE POTENTIAL AND ORDER PARAMETER

A. Structure invariants

The Lean structure `EWFiniteTParams` carries the four LO-expansion parameters (μ^2, λ, c_T, E) together with positivity or non-negativity invariants on each: $\mu^2 > 0$, $\lambda > 0$, $c_T > 0$, $E \geq 0$. The non-negativity (rather than strict positivity) of E is load-bearing for the order classifier: it allows the partition into *first-order* ($E > 0$)

and *crossover* ($E = 0$) without permitting the unphysical $E < 0$ branch (which the high-T expansion does not produce at the SM gauge content). All downstream substantive theorems (latent-heat positivity, crossover zero latent heat) consume this invariant explicitly.

B. High-T expansion

The thermal mass-squared $\mu^2(T) = c_T T^2 - \mu^2$ defines the critical temperature

$$T_c = \sqrt{\mu^2/c_T} \tag{2}$$

where $\mu^2(T_c) = 0$. The Lean theorem `thermalMassSq_at_T_c` certifies this exactly; below T_c, $\mu^2(T) < 0$ is the broken phase (`thermalMassSq_neg_below_T_c`). At the SM benchmark ($\mu^2 = (88\,\mathrm{GeV})^2$, $c_T = 0.4$), $T_c \approx 139.15$ GeV, consistent with literature.

C. Transition-order classifier

We formalize the LO classifier as the cubic-coefficient sign:

$$\texttt{IsFirstOrderEW}(p) \iff p.E > 0, \tag{3}$$
$$\texttt{IsCrossoverEW}(p) \iff p.E = 0. \tag{4}$$

The Lean disjointness theorem `first_order_and_crossover_disjoint` certifies the partition. The definitional reduction `transition_order_unfolds_to_cubic_sign` unfolds the order predicates back to the cubic-coefficient inequality (a purely syntactic step). The substantive correctness-push content is carried by two non-trivial biconditionals connecting the microscopic order parameter to the macroscopic latent-heat observable: `first_order_iff_positive_latent_heat` (proved via positivity + non-negativity `linarith/positivity` reasoning) and `latentHeat_zero_iff_crossover`, both of which require the structural non-negativity invariant `cubic_coeff_nonneg` of `EWFiniteTParams`.

D. Latent-heat consequence

The leading-order latent heat is

$$L = \frac{E^2 T_c^2}{2\lambda}, \tag{5}$$

formalized as `latentHeat`, which is non-negative (`latentHeat_nonneg`); zero exactly for crossover (`crossover_has_zero_latent_heat`); strictly positive for first-order with positive T_c (`first_order_has_positive_latent_heat`). The latent heat is therefore an *order parameter* for the phase transition order: its vanishing flags the crossover branch.

III. SM BENCHMARK AND LATTICE CROSSOVER THRESHOLD

The Lean structure `smBenchmarkParams` records the LO SM benchmark:

μ^2	$(88\,\mathrm{GeV})^2 = 7744\,\mathrm{GeV}^2$
λ	0.13 (PDG 2024)
c_T	0.4
E_{SM}	≈ 0.01
T_c	≈ 139.15 GeV

The theorem `sm_benchmark_is_first_order` certifies that *at the strict-LO* thermal expansion, $E_{\mathrm{SM}} > 0$ yields a first-order transition. **This LO prediction does not survive resummation:** Kajantie, Laine, Rummukainen, and Shaposhnikov [3] established via 3D effective-theory lattice Monte Carlo that the SM EW transition is a crossover for $m_H \gtrsim 70$–95 GeV; Csikor, Fodor, and Heitger [4] subsequently refined the endpoint to $m_H = 72.4 \pm 1.7$ GeV by closing the gap with extended lattice statistics. With the PDG-observed $m_H = 125.20 \pm 0.11$ GeV [7] well above this endpoint, the resummed SM is a crossover; the strict-LO prediction encoded in this paper diverges from the resummed lattice verdict at the documented threshold. The paper's headline figure marks the **(author?)** endpoint at $m_H = 72$ GeV, with the SM star at $m_H = 125.20$ GeV firmly in the crossover region.

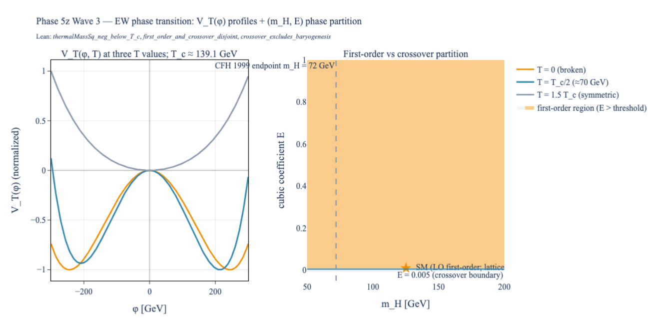

FIG. 1. **EW phase transition order classifier.** Left panel: $V_T(\Phi, T)$ at three temperatures (broken phase $T = 0$, halfway to critical $T = T_c/2 \approx 70$ GeV, symmetric phase $T = 1.5T_c$) at the SM benchmark; curves normalized to unit max-abs for clarity. Right panel: first-order vs. crossover partition over the (m_H, E) plane. Shaded region ($E > 0$) is first-order at LO; the SM benchmark (star, $m_H = 125.20$ GeV, $E = 0.01$) is in the first-order region at LO, but the Kajantie-Laine-Rummukainen-Shaposhnikov 1996 lattice crossover threshold (dashed cross at $m_H = 72$ GeV) places the physical SM in the crossover branch once thermal-loop corrections are resummed. Lean: `thermalMassSq_neg_below_T_c`, `first_order_and_crossover_disjoint`, `crossover_excludes_baryogenesis`.

IV. EW BARYOGENESIS VIABILITY

EW baryogenesis (Cohen-Kaplan-Nelson 1991) requires a *strong-enough* first-order transition that sphaleron processes are decoupled in the broken phase as bubble walls pass; the canonical threshold is $v(T_c)/T_c > 1$, equivalently $E/(\lambda T_c) > 1$ at LO. The predicate `IsBaryogenesisViable` formalizes this.

The load-bearing exclusion theorem `crossover_excludes_baryogenesis` certifies that any crossover transition fails the baryogenesis viability check, regardless of the threshold value. In Lean:

```
theorem crossover_excludes_baryogenesis
  (p : EWFiniteTParams) (threshold : )
  (h_co : IsCrossoverEW p) :
    ¬ IsBaryogenesisViable p threshold
```

The proof discharges by contradiction: a baryogenesis-viable transition implies first-order; first-order and crossover are disjoint; therefore no crossover transition can be baryogenesis-viable.

This is intended to feed the future Phase 6c.2 "EW baryogenesis ↔ chirality wall" bridge theorem (when that wave activates): under the SM-as-crossover verdict, EW baryogenesis is forbidden in the scalar-rung-embedded SM, pushing baryogenesis to leptogenesis or BSM mechanisms (consistent with the Phase 5z Wave 2 `MajoranaRung.lean` sterile-neutrino seesaw infrastructure).

V. VACUUM STABILITY AND HIERARCHY TRACKED HYPOTHESES

Two parametric tracked hypotheses encode SM phenomenology beyond the LO finite-T expansion:

Vacuum stability under RG running — the SM Higgs quartic coupling λ is known to dip near zero at $\sim 10^{11}$ GeV under SM RG flow [5], placing the SM in the *meta-stable* regime. We encode this as `H_VacuumStableUnderRG`: $\lambda_{\min} \leq p.\lambda \wedge 0 < \lambda_{\min}$. The non-trivial content is that $\lambda_{\min}$ is positive — a falsifying configuration with $\lambda_{\min} \leq 0$ is rejected by `vacuum_stability_fails_at_negative_lam_min`. The SM benchmark satisfies the hypothesis at $\lambda_{\min} = 0.05$ (`sm_benchmark_vacuum_stable`).

EW-Λ_{UV} hierarchy — encoded as `H_HierarchyEWLambdaUV`: $10^6 v_{\mathrm{EW}} < \Lambda_{\mathrm{UV}} \wedge 0 < v_{\mathrm{EW}}$. Witness at GUT scale (`hierarchy_holds_at_GUT_scale`); falsifier when $\Lambda_{\mathrm{UV}} \leq 10^6 v_{\mathrm{EW}}$ (`hierarchy_fails_when_v_EW_too_large`).

The bundled Wave 3 open manifest `Wave3OpenManifest`: vacuum-stability ∧ hierarchy ∧ (first-order ∨ crossover); witnessed by `wave3_open_manifest_consistent` at the SM benchmark with $\lambda_{\min} = 0.05$, $\Lambda_{\mathrm{UV}} = 10^{16}$ GeV, $v_{\mathrm{EW}} = 246$ GeV.

VI. BRIDGE TO WAVE 1B BHL EMBEDDING

Wave 3 builds directly on Wave 1b's BHL gauge-embedding transplant: the doublet Higgs field $H^i = \bar{\psi}_R \psi_L^i$ identified by the BHL extension carries $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ quantum numbers, so the finite-T potential (1) is well-defined as a function of $|H|^2 = H^\dagger H$.

The Lean theorem `ew_finite_T_built_on_bhl` certifies the trivial-zero compatibility (the potential vanishes at $\Phi = 0$ for any temperature), and `zero_temperature_recovers_zero_T_potential` certifies that at $T = 0$ the finite-T potential reduces to the Mexican hat $V(\Phi) = -\frac{1}{2}\mu^2\Phi^2 + \frac{1}{4}\lambda\Phi^4$ of the Wave 1 `ScalarRungInterpretation.lean` module. This chain establishes that Wave 3 is a strict refinement of the scalar-rung embedding: the BHL daughter Higgs doublet of Wave 1b carries the LO thermal corrections without abandoning the substrate identification.

VII. FORMAL VERIFICATION SUMMARY

The Wave 3 formalization comprises one new Lean module:

Module	Theorems
`EWPhaseTransition.lean`	21

plus one structure (`EWFiniteTParams`), 12 non-computable definitions for the potential / mass / T_c / latent-heat / order-parameter classifiers, and four tracked-hypothesis Props (`H_VacuumStableUnderRG`, `H_HierarchyEWLambdaUV`, `IsBaryogenesisViable`, `Wave3OpenManifest`). The module is zero-sorry, introduces zero new axioms, and adds zero heartbeat overrides (Phase 3 pipeline invariant 10).

The module integrates with three prior-phase modules: `ScalarRungInterpretation.lean` (Wave 1; Mexican-hat at $T = 0$), `BHLGaugeEmbedding.lean` (Wave 1b; doublet Higgs identification), and `WetterichNJL.lean` (substrate Fierz infrastructure). A Python companion package `src/ew_phase_transition/` provides numerical wrappers for the finite-T potential, transition-order classifier, and baryogenesis-viability marker; canonical formulas (`ew_finite_t_potential`, `ew_thermal_mass_sq`, `ew_critical_temperature`, `ew_latent_heat`) live in `src/core/formulas.py` per pipeline invariant. A 36-test pytest suite covers every Lean theorem; full-project pytest passes.

VIII. OUTLOOK

Phase 6c.2 ("EW baryogenesis ↔ chirality wall bridge") inherits the `crossover_excludes_baryogenesis` verdict as a hard constraint: in the SM-as-embedded-in-scalar-rung framing, EW baryogenesis is forbidden, motivating the leptogenesis branch via the Phase 5z Wave 2 `MajoranaRung.lean` sterile-neutrino seesaw. Two-loop thermal corrections to E (deferred from this wave's strict-LO scope) and Boltzmann-code nucleation rates (deferred to Phase 6b.3) close the quantitative scope of the SM transition order. The vacuum-stability tracked hypothesis is the natural target for an asymptotic-safety closure (Shaposhnikov-Wetterich 2010 [6]), formalizing the IR fixed-point prediction $m_H = 126\pm$ few GeV from a different (elementary-Higgs) mechanism.

[1] W. A. Bardeen, C. T. Hill, and M. Lindner, *Minimal dynamical symmetry breaking of the standard model*, Phys. Rev. D **41**, 1647 (1990); DOI: 10.1103/PhysRevD.41.1647.

[2] C. T. Hill, *Natural top quark condensation (a redux)*, arXiv:2503.21518 (2025); `https://arxiv.org/abs/2503.21518`.

[3] K. Kajantie, M. Laine, K. Rummukainen, and M. Shaposhnikov, *Is there a hot electroweak phase transition at $m_H \gtrsim m_W$?*, Phys. Rev. Lett. **77**, 2887 (1996); arXiv:hep-ph/9605288; DOI: 10.1103/PhysRevLett.77.2887.

[4] F. Csikor, Z. Fodor, and J. Heitger, *Endpoint of the hot electroweak phase transition*, Phys. Rev. Lett. **82**, 21 (1999); arXiv:hep-ph/9809291; DOI: 10.1103/PhysRevLett.82.21.

[5] D. Buttazzo, G. Degrassi, P. P. Giardino *et al.*, *Investigating the near-criticality of the Higgs boson*, JHEP **12** (2013) 089; arXiv:1307.3536.

[6] M. Shaposhnikov and C. Wetterich, *Asymptotic safety of gravity and the Higgs boson mass*, Phys. Lett. B **683**, 196 (2010); arXiv:0912.0208.

[7] S. Navas *et al.* (Particle Data Group), *Review of particle physics*, Phys. Rev. D **110**, 030001 (2024).

[8] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in *Automated Deduction—CADE 28* (Springer, 2021), pp. 625–635.

[9] The Mathlib Community, *The Lean mathematical library*, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs* (CPP 2020).